

# Distributional Semantic Models

## Part 4: A linguistic perspective on DSMs

Stephanie Evert<sup>1</sup>

with Gabriella Lapesa<sup>2</sup>, Alessandro Lenci<sup>3</sup>, and Marco Baroni<sup>4</sup>

<sup>1</sup>Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

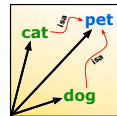
<sup>2</sup>Heinrich-Heine-Universität Düsseldorf & GESIS, Germany

<sup>3</sup>University of Pisa, Italy

<sup>4</sup>ICREA / Universitat Pompeu Fabra Barcelona, Italy

<http://wordspace.collocations.de/doku.php/course:start>

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## Outline

Does evaluation tell us everything?

A cautionary tale

Beyond NLP: Linguistic evaluation

Polysemy

Compositionality

Other linguistic aspects

The FAST task

Free association norms

FAST: Data set and tasks

FAST: Experiments

Hands-on exercises

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## The CogALex-V Shared Task

Santus *et al.* (2016)

- ▶ Aim: better linguistic understanding of DS from identification of specific **semantic relations**
- ▶ Data: 747 target words with approx. 10 candidate relata each
  - ▶ training set: 318 targets, 3054 word pairs
  - ▶ test set: 429 targets, 4260 word pairs
- ▶ Subtask 1: related *vs.* unrelated word pairs
  - ▶ unrelated pairs are random fillers
  - ▶ relatively easy:  $F_1 = 79.0\%$  (best system)
- ▶ Subtask 2: distinguish between semantic relations
  - ▶ SYN:  $w_2$  can be used with same meaning as  $w_1$
  - ▶ ANT:  $w_2$  can be used as the opposite of  $w_1$
  - ▶ HYPER:  $w_1$  is a kind of  $w_2$
  - ▶ PART\_OF:  $w_1$  is a part of  $w_2$
  - ▶ RANDOM: no relation (random word + manual check)
  - ▶ relatively hard:  $F_1 = 44.5\%$  (best system: **deep learning**)

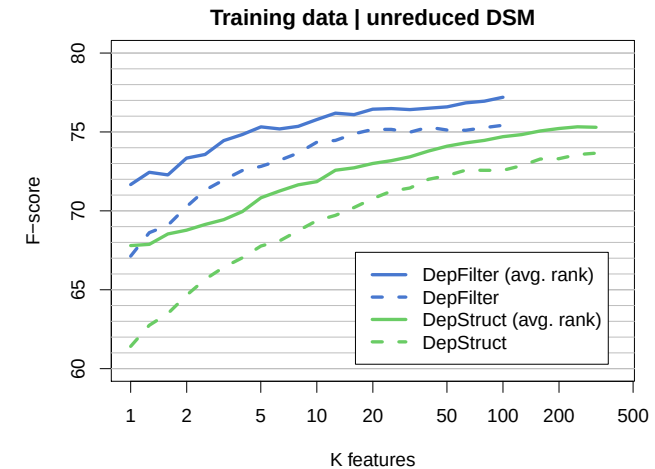
## Mach 5 at CogALex 2016

Evert (2016)

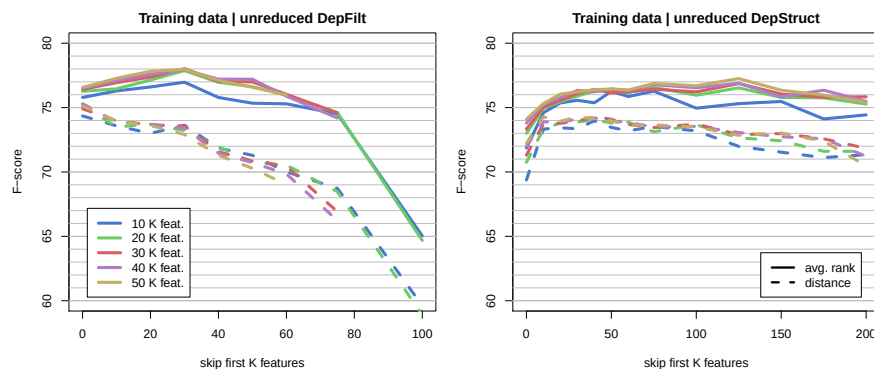
- ▶ Mach 5 participated in the CogALex-V Shared Task as a traditional “count” (non-neural) DSM
  - ▶ 10-billion-word Web corpus (Schäfer & Bildhauer 2012)
  - ▶ syntactic dependencies from C&C parser (Curran *et al.* 2007)
  - ▶ 26.5k target words, up to 300k feature dimensions
  - ▶ other parameters set according to Lapesa & Evert (2014)
- ▶ Parameter optimization on training data (subtask 1)
- ▶ Machine learning on optimized representations (subtask 2)
  - ▶ learns relevance weights for 600 latent SVD dimensions
  - ▶ best results from combination of different SVD spaces

Try it yourself: <http://www.collocations.de/data/#mach5>

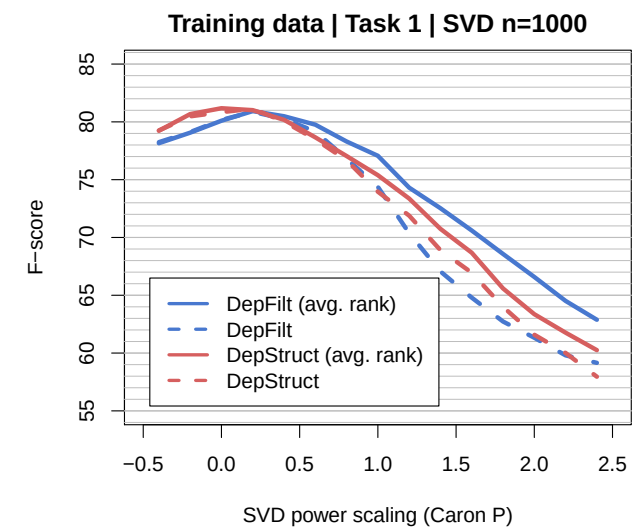
## Mach 5: Parameter optimization



## Mach 5: Parameter optimization



## Mach 5: Parameter optimization



## Mach 5: Are we doing well?

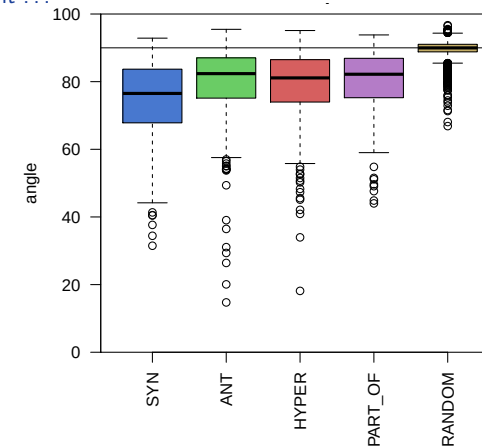
$F_1 = 77.88\%$  for related vs. unrelated (best: 79.0%)

However ...

- ▶ Parameter optimization yields surprising result:  
best model uses < 50k features with relatively low frequency
- ▶ Nearest neighbours are unsatisfactory, e.g. for *play*:  
*playing* (54.1°), *star* (62.8°), *reunite* (62.9°), *co-star* (64.3°),  
*reprise* (64.4°), *player* (66.7°), *score* (68.5°), *audition* (69.2°),  
*sing* (69.4°), *actor* (69.5), *understudy* (69.6), *game* (70.3), ...
- ▶ Why is Mach 5 still doing so well in the task, then?

## Mach 5: What is going wrong?

A disturbing result ...



- DSM has learned to recognize random word pairs (at 90°)!
- We need better data sets with **high-quality distractors**!

## Problems of standard tasks & data sets

Problems with semantic interpretation of DSMs don't only stem from evaluation methodology ...

... **data sets can be problematic as well!**

Two major problems:

- ▶ DSMs may exploit contingent properties of the task
  - ▶ **random fillers** as distractors ("controls")
    - ➔ recognize random word pairs rather than semantic relations
  - ▶ choice of clearly separated categories and prototypical exemplars in noun clustering task (ESSLI 2008)
    - ➔ much harder to identify categories in general word list
  - ▶ typical superordinate-level words in hypernym detection task
    - ➔ recognize "typical hypernym" in a multiple-choice setting
- ▶ Data set size too small
  - ▶ e.g. 97.5% accuracy on 80 TOEFL items ➔ over-fitting

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## DSM similarity & linguistic theory

Numerical results from evaluation tasks can be misleading ...

📖 **perhaps a linguistic perspective gives us better insights?**

1. Polysemy
  - ▶ a textbook challenge, will look at most intuitive solution
  - 📖 hands-on: [hands\\_on\\_part4\\_schuetze1998.R](#)
2. Compositionality
  - ▶ above and below the word level
  - 📖 hands-on: derivational morphology (Lazaridou *et al.* 2013)
3. Not all meaning is distributional
  - ▶ e.g. function words, proper names
4. Distributional vectors as collocational profiles

Great overview paper:

[Distributional Semantics and Linguistic Theory](#) (Boleda 2020)

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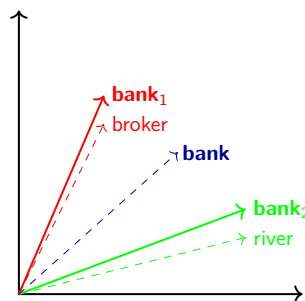
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## Polysemy in DSMs

- ▶ **Problem:** DSM vectors conflate contexts from different senses
  - ▶ collocates of “bank”: *money, river, account, swim, ...*
  - ▶ conflated DSM vector either fits neither meaning well or is dominated by most frequent sense



## Polysemy in DSMs

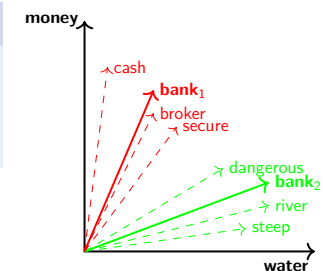
Observation: DSM vectors conflate contexts from word senses

- ▶ **Solution:** build a representation for each instance of the word we want to disambiguate (Schütze 1998)

📖 sentence vectors

Target: bank

**bank<sub>1</sub>:** *The broker went to the bank to secure his cash*  
**bank<sub>2</sub>:** *The river bank was steep and dangerous*



Application: word sense disambiguation

... can you think about another situation in which we may need it?

Context vectors: can we do them in **wordspace**?

Yes :-)

```
library(wordspace)
# S1: "Cats and dogs need their time"
s1 <- "cat and dog need their time"
# S2: "Time is the cause not the effect"
s2 <- "time is the cause not the effect"
# Ingredients: vectors for individual words
>TT <- DSM_TermTermMatrix
>TT
      breed tail feed kill important explain likely
cat      84   17    8   38         0         2     0
dog     579   14   32   63         1         2     2
animal   45   11   86  136        13         5     4
time     19    8   29  134        94        44    100
reason    1    0    1   18        71       140    39
cause     0    1    0    3        55        35    51
effect    0    1    1    6        62        37    14
```

Context vectors: can we do the, in **wordspace**?

Yes :D

“cats and dogs need their time”

```
> context.vectors(TT, s1)
      breed tail feed      kill important explain likely
1 227.3333   13   23 78.33333 31.66667      16      34
# context.vectors() is taking the average of the values in each cell
> (TT['cat', 'breed'] + TT['dog', 'breed'] + TT['time', 'breed'])/3
227.3333
```

“time is the cause not the effect”

```
round(context.vectors(TT, s2), 3)
      breed tail feed      kill important explain likely
1 6.333 3.333   10 47.667   70.333 38.667      55
```

Context vectors: can we do them in **wordspace**?

Almost there ...

```
# context.vectors() can also take a list as an input
contexts <- round(context.vectors(TT, c(s1, s2)), 2)
# The output is a matrix, let's give it better rownames first
rownames(contexts) <- c("s1", "s2")
# ... and then append it to our original matrix
TT <- rbind(TT, contexts)
TT
      breed tail feed      kill important explain likely
cat      84.00 17.00    8 38.00         0.00    2.00     0
dog     579.00 14.00   32 63.00         1.00    2.00     2
animal   45.00 11.00   86 136.00        13.00    5.00     4
time     19.00  8.00   29 134.00        94.00   44.00    100
reason    1.00  0.00    1 18.00        71.00  140.00    39
cause     0.00  1.00    0  3.00        55.00   35.00    51
effect    0.00  1.00    1  6.00        62.00   37.00    14
s1      227.33 13.00   23 78.33        31.67   16.00    34
s2       6.33  3.33   10 47.67        70.33   38.67    55
```

Context vectors: can we do them in **wordspace**?

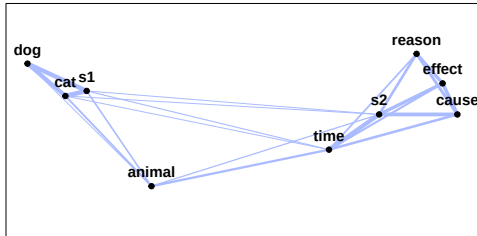
And what now?

```
# We can do all the cool things we are used to do with DSM matrices
# Nearest neighbors...
nearest.neighbours(TT, c("s1", "s2"), n=6)
$s1
      cat      dog  animal      time      s2      cause
14.31016 17.16200 55.27587 62.66470 67.81707 77.90557

$s2
      time      cause      effect      reason      animal      s1
18.85097 25.19348 31.51682 40.83768 60.61621 67.81707
```

## Context vectors: can we do them in **wordspace**?

```
# And a semantic map!
plot(dist.matrix(TT))
```



`hands_on_part4.R` also contains an example for *bank* with `word2vec` embeddings. If you enjoy centroids, try `hands_on_part4_schuetze1998.R`!

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## Compositionality

Can we capture it in DS?

- Formally: compositionality implies some operator  $\oplus$  such that
 
$$\text{meaning}(w_1 w_2) = \text{meaning}(w_1) \oplus \text{meaning}(w_2)$$
- CDSM recipe
  - **Distributional vectors**  $\mathbf{v}_1$  for  $w_1$  and  $\mathbf{v}_2$  for  $w_2$
  - **Operators**: mathematical strategies to combine  $\mathbf{v}_1$  and  $\mathbf{v}_2$  to predict a vector representation  $\mathbf{v}_1 \oplus \mathbf{v}_2$  for  $w_1 w_2$ 
    - ★ vector addition
    - ★ vector multiplication
    - ★ nonlinear operations learned by neural networks
- Problem: some words (e.g. *not*) are themselves more like operators than points in space

Great overview paper: [Frege in space: A program for compositional distributional semantics](#) (Baroni et al. 2014)

## Compositionality with distributional vectors

Additive and Multiplicative Models (Mitchell & Lapata 2010)

|            | music | solution | economy | craft | create |
|------------|-------|----------|---------|-------|--------|
| practical  | 0     | 6        | 2       | 10    | 4      |
| difficulty | 1     | 8        | 4       | 4     | 0      |
| problem    | 2     | 15       | 7       | 9     | 1      |

$$\mathbf{w} = \mathbf{u} + \mathbf{v}$$

`predicted(practical difficulty) = practical + difficulty = [1 14 6 14 4]`

$$\mathbf{w} = \mathbf{u} \odot \mathbf{v}$$

`predicted(practical difficulty) = practical  $\odot$  difficulty = [0 48 8 40 0]`

What is your intuition about the effect of multiplication? How is it different from addition? Have you already seen it as an ingredient of something else?

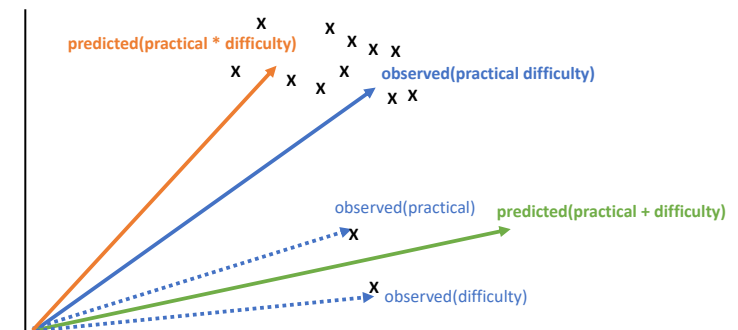
## How do I know my compositional vectors are “good”?

Evaluation, again :)

- Qualitative inspection** of nearest neighbors of  $\mathbf{u} \oplus \mathbf{v}$ 
  - In which case look the neighbours intuitively more sensible?
  - problem* would be “good” neighbour of *practical*  $\oplus$  *difficulty*
- Quantitative evaluation**
  - Collect DSM vector for *practical difficulty* in the same corpus
  - Evaluation criterion:  
observed(*practical difficulty*)  $\approx$  predicted(*practical difficulty*)
    - ★ Would you expect  $+$  or  $\odot$  to work better?
  - Evaluation metric
    - ★ distance(predicted, observed) (Lazaridou *et al.* 2013)
    - ★ neighbour rank (Baroni & Zamparelli 2010; Padó *et al.* 2016)

## How do I know my compositional vectors are “good”?

Observed vs. predicted vector



rank(predicted(*practical* + *difficulty*)) = 5 < rank(predicted(*practical* \* *difficulty*)) = 10

distance(predicted(*practical* \* *difficulty*)) < distance(predicted(*practical* + *difficulty*))

## Adjective-noun composition (Baroni & Zamparelli 2010)

Starting point: observed AN vectors

- Input: triples of {observed(AN), A, N}
  - {bad luck, bad, luck}, {red cover, red, cover}, etc.
  - 36 adjectives (size, color, temporal, etc.)

|                        |                               |                       |
|------------------------|-------------------------------|-----------------------|
| <i>bad luck</i>        | <i>electronic communities</i> | <i>historical map</i> |
| bad                    | electronic storage            | topographical         |
| bad weekend            | electronic transmission       | atlas                 |
| good spirit            | purpose                       | historical material   |
| <i>important route</i> | <i>nice girl</i>              | <i>little war</i>     |
| important transport    | good girl                     | great war             |
| important road         | big girl                      | major war             |
| major road             | guy                           | small war             |
| <i>red cover</i>       | <i>special collection</i>     | <i>young husband</i>  |
| black cover            | general collection            | small son             |
| hardback               | small collection              | small daughter        |
| red label              | archives                      | mistress              |

- Methods: increasing computational complexity
  - unsupervised (additive, multiplicative)
  - supervised: learns matrix *A* by comparing AN and N

## Adjective-noun composition (Baroni & Zamparelli 2010)

Observed(AN) vs. predicted(AN): neighbours

| SIMILAR                 |                      |                       | DISSIMILAR        |                      |                       |
|-------------------------|----------------------|-----------------------|-------------------|----------------------|-----------------------|
| <i>adj N</i>            | <i>obs. neighbor</i> | <i>pred. neighbor</i> | <i>adj N</i>      | <i>obs. neighbor</i> | <i>pred. neighbor</i> |
| common understanding    | common approach      | common vision         | American affair   | Am. development      | Am. policy            |
| different authority     | diff. objective      | diff. description     | current dimension | left (a)             | current element       |
| different partner       | diff. organisation   | diff. department      | good complaint    | current complaint    | good beginning        |
| general question        | general issue        | <i>same</i>           | great field       | excellent field      | gr. distribution      |
| historical introduction | hist. background     | <i>same</i>           | historical thing  | different today      | hist. reality         |
| necessary qualification | nec. experience      | <i>same</i>           | important summer  | summer               | big holiday           |
| new actor               | new cast             | <i>same</i>           | large pass        | historical region    | large dimension       |
| recent request          | recent enquiry       | <i>same</i>           | special something | little animal        | special thing         |
| small drop              | droplet              | drop                  | white profile     | chrome (n)           | white show            |
| young engineer          | young designer       | y. engineering        | young photo       | important song       | young image           |

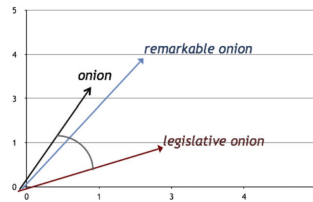
Table 4: Left: nearest neighbors of observed and *alm*-predicted ANs (excluding each other) for a random set of ANs where rank of observed w.r.t. predicted is 1. Right: nearest neighbors of predicted and observed ANs for random set where rank of observed w.r.t. predicted is  $\geq 1K$ .

## How about unattested AN combinations?

Capturing semantically deviant AN combinations (Vecchi *et al.* 2017)

**Can we use compositional DSMs to tell, among equally unattested AN, which one is semantically less plausible?**

The *composed vectors* for semantically deviant combinations (human ratings) will be **farther away** from the head noun than the acceptable ones



... they test other measures (e.g., neighbors density, vector length) as well as different composition methods: have a look at the paper!

## How about unattested AN combinations?

Capturing semantically deviant AN combinations (Vecchi *et al.* 2017)

**Can we use compositional DSMs to tell, among equally unattested AN, which one is semantically less plausible?**

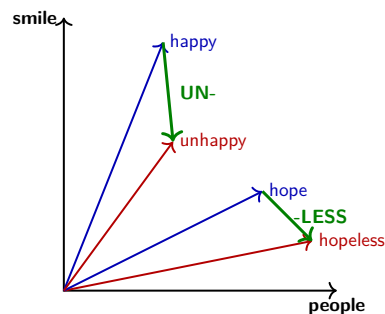
**Qualitative inspection:** the *composed vectors* of semantically acceptable pairs have plausible nearest neighbors

- a. \*angry lamp      { *shocked, fearful, angry, defiant* }
- b. \*nuclear fox     { *nuclear, nuclear arm, nuclear development, nuclear expert* }
- c. warm garlic     { *green salad, wild mushroom, sauce, green sauce* }
- d. spectacular striker { *goal, crucial goal, famous goal, amazing goal* }

`hands_on_part4.R` contains an implementation of vector addition and multiplication in wordspace. Have fun chasing the strangest AN combinations (and others as well)!

## Compositionality below the word level

Can we use compositional DSMs to investigate the meaning of derivational patterns?



- ▶ Starting point: vectors for **base** and **derived** words.
- ▶ Two strategies:
  - ✎ learn the **semantic shifts** with compositional methods
  - ▶ investigate **properties** of the **patterns** → semantic relations
    - ★ zero-nominalisations as hyponyms of the base verb (Varvara *et al.* 2021)
    - ★ *un-* as antonyms of the base nouns

## The DS of Derivational Morphology (Lazaridou *et al.* 2013)

1. Input: derived/stem vector pairs for each affix
  - ▶ *un-*: unfaithful/fairful, unbiased/biased, unwell/well
  - ▶ *-ly*: true/truly, mad/madly, deep/deeply
2. Goal: **build one representation per affix**
  - ▶ No (well, little) learning (additive and multiplicative)
    - ★ *un-* = centroid(unfaithful, unbiased, unwell, etc.)
  - ▶ Increasingly complex learning
    - ★ parameters set during training to optimize composition, affixes as matrices (cf. adjectives)
3. Prediction & evaluation
  - ▶ Apply affix to unseen base and compare predicted(derived) *vs.* observed(derived)
    - ★ simplest (additive) & most complex (lexical functional, theoretically motivated): comparable
    - ★ cf. Padó *et al.* (2016) for German: simple composition methods work better!

## The DS of Derivational Morphology (Lazaridou *et al.* 2013)

### Dataset

| Affix      | Stem/Der. POS | Training Items | HQ/Tot. Test Items | Avg. SDR |
|------------|---------------|----------------|--------------------|----------|
| -able      | verb/adj      | 177            | 30/50              | 5.96     |
| -al        | noun/adj      | 245            | 41/50              | 5.88     |
| -er        | verb/noun     | 824            | 33/50              | 5.51     |
| -ful       | noun/adj      | 53             | 42/50              | 6.11     |
| -ic        | noun/adj      | 280            | 43/50              | 5.99     |
| -ion       | verb/noun     | 637            | 38/50              | 6.22     |
| -ist       | noun/noun     | 244            | 38/50              | 6.16     |
| -ity       | adj/noun      | 372            | 33/50              | 6.19     |
| -ize       | noun/verb     | 105            | 40/50              | 5.96     |
| -less      | noun/adj      | 122            | 35/50              | 3.72     |
| -ly        | adj/adv       | 1847           | 20/50              | 6.33     |
| -ment      | verb/noun     | 165            | 38/50              | 6.06     |
| -ness      | adj/noun      | 602            | 33/50              | 6.29     |
| -ous       | noun/adj      | 157            | 35/50              | 5.94     |
| -y         | noun/adj      | 404            | 27/50              | 5.25     |
| in-        | adj/adj       | 101            | 34/50              | 3.39     |
| re-        | verb/verb     | 86             | 27/50              | 5.28     |
| un-        | adj/adj       | 128            | 36/50              | 3.23     |
| <i>tot</i> | <i>*/*</i>    | 6549           | 623/900            | 5.52     |

7000 base/derived pairs from CELEX, 18 patterns, training vs. test (further annotated for base/derived relatedness and vector quality)

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## Not all semantic knowledge is distributional

**Proper names** “answer the purpose of **showing** what thing it is that we are talking about but not of telling anything about it” (Mill, 1843)

- ▶ Intuition: instances of categories such as PER, ORG, etc.
- ▶ Herbelot (2015), standard DSMs: category → instance
  - ▶ “... upon encountering the name *Mr Darcy* for the first time in the novel, a reader will attribute it the representation of the concept MAN and subsequently **specialise** it as per the linguistic contexts in which the name appears”
- ▶ Westera *et al.* (2021), embeddings: instance → category

**Function words**: some pointers

- ▶ Baroni *et al.* (2012) on quantifiers/entailment, Bernardi *et al.* (2013) on determiners, Padó & Hole (2022) on the polysemy of the German reflexive *sich*

## Making sense: collocational profiles vs. DSM vectors

Some exciting open questions for your consideration

- ▶ For an unreduced term-term DSM, each row vector  $\mathbf{m}_t$  corresponds to the collocational profile of a target term  $t$ 
  - ▶ e.g. dependency-structured DSM → word sketches
  - ▶ compare our own reading of sketch differences for nearest neighbours with distributional similarity
- ▶ Can dim.-reduced vectors be read as collocational profiles?
  - ▶ map to original features → dense “virtual” collocational profile
  - ▶ compare top- $k$  entries with original sparse profile
- ▶ Do DSM distances use the same information as the human interpretation of collocational profiles?
  - ▶ humans likely to focus on few salient collocates, while DSM might exploit weak correlations across many dimensions
  - ▶ cf. insights from authorship attribution (Evert *et al.* 2017)

## Making sense: collocational profiles vs. DSM vectors

Some exciting open questions for your consideration

- ▶ Can we figure out which features account for semantic differences?
  - ▶ normalise vectors:  $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$
  - ▶  $(u_i - v_i)^2$  = contribution to (squared) Euclidean distance
  - ▶ for cosine similarity, contribution is  $u_i v_i$
- ▶ Collocations = **first-order** association = **syntagmatic**
- ▶ DSM similarity = **second-order** association = **paradigmatic**
  - ▶ widely-held belief in distributional semantics, but is it true?
  - ▶ some applications actually require both first-order and second-order association (e.g. **free association norms**)
  - ▶ we will obtain a few mathematical insights in Part 5

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## Cognitive modelling with DSMs

- ▶ **Why?** – Because we want to know whether DS captures the mental lexical knowledge of human speakers!
- ▶ Task: DSM predicts reaction times in **priming experiments** (Hare *et al.* 2009; Lapesa & Evert 2013)
  - ▶ often just experimental items used for multiple-choice task (e.g. Padó & Lapata 2007; Herdağdelen *et al.* 2009)
  - ▶ cf. tasks constructed from Lazaridou2013 in previous section
  - ▶ data sets of experimental items: **GEK\_Items**, **SPP\_Items**
- ▶ Task: DSM predicts **EEG potentials** (Murphy *et al.* 2009) or **fMRI brain activation** levels (Mitchell *et al.* 2008)
  - ▶ huge datasets, but tiny and selective vocabulary
- ▶ Task: DSM predicts human **free associations**
  - ▶ often considered a “window into the mental lexicon”
  - ▶ free association norms available for thousands of cue words

## Free associations

... a cue into the organization of the mental lexicon?

Which words come to your mind if you hear ...

- ▶ whisky → gin, drink, scotch, bottle, soda
- ▶ giraffe → neck, animal, zoo, long, tall

- ▶ Hypotheses concerning the nature of the underlying process:
  - ▶ Result of learning by contiguity (James 1890) **syntagmatic (1<sup>st</sup>-order)**
  - ▶ Result of symbolic processes which make use of complex semantic structures (Clark 1970) **paradigmatic (2<sup>nd</sup>-order)**
- ▶ Large collections available
  - ▶ Edinburgh Associative Thesaurus (**EAT**)  
8210 stimuli, 100 subjects (Kiss *et al.* 1973)
  - ▶ University of South Florida Free Association Norms (**USF**)  
5019 stimuli, 6000 subjects (Nelson *et al.* 2004)

## Syntagmatic vs. paradigmatic relations

### Definitions and general assumptions

- ▶ **Syntagmatic**  $\longleftrightarrow$  contiguity
  - ▶ Examples: {dog, barks}, {dog, bone}
  - ▶ Words appear together: 1<sup>st</sup>-order co-occurrence
  - ▶ Found in: collocational profiles, DSM dimensions
- ▶ **Paradigmatic**  $\longleftrightarrow$  interchangeability
  - ▶ Examples: {book, volume}, {dog, animal}
  - ▶ Words appear in similar contexts: 2<sup>nd</sup>-order co-occurrence
  - ▶ Usually semantically related
  - ▶ Found in: DSM nearest neighbours

### However ...

DSM neighbourhoods include syntagmatically related words (collocates) if certain parameters are properly set, in particular if the context window is large enough (Lapesa *et al.* 2014).

## Free associations in a DSM

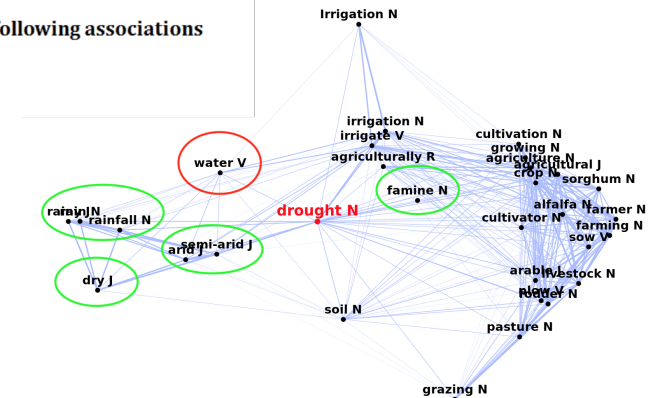
### Drought in EAT vs. DSM

#### drought stimulated the following associations

Number of different answers: 30

Total count of all answers: 97

- WATER 21 0.22
- DRY 16 0.16
- THIRST 9 0.09
- FAMINE 7 0.07
- RAIN 7 0.07
- DESERT 6 0.06
- BEER 5 0.05
- CRACK 2 0.02
- HOT 2 0.02
- SAND 2 0.02
- ALE 1 0.01
- ARID 1 0.01
- AUSTRALIA 1 0.01
- CATTLE 1 0.01
- COLD 1 0.01
- COOL 1 0.01
- DEATH 1 0.01
- DUST 1 0.01
- GALE 1 0.01
- MONSOON 1 0.01



## Free associations & co-occurrence data

### Previous work

- ▶ Wettler *et al.* (2005)
  - ▶ Data: subset of EAT (100 stimuli)
  - ▶ Task: prediction of the most common free associate
  - ▶ Model: first-order model, BNC, large window (20 words)
  - ▶ Result: human associative responses can be predicted from contiguities between words in language use (collocations)
- ▶ ESSLLI 2008 Shared Task
  - ▶ Data: subset of EAT (a different set of 100 stimuli)
  - ▶ Task 1: discrimination btw. the most common associate and hapax/random distractors → multiple choice
  - ▶ Task 2: prediction of the most common free associate
  - ▶ Result: first-order models (collocations) are better than second-order models (DSMs)

## Outline

### Does evaluation tell us everything?

A cautionary tale

### Beyond NLP: Linguistic evaluation

Polysemy  
Compositionality  
Other linguistic aspects

### The FAST task

Free association norms  
FAST: Data set and tasks  
FAST: Experiments  
Hands-on exercises

## The Free ASsociation Task (FAST) data set

### Preprocessing

1. Starting point: EAT (8210 stimuli), USF (5019 stimuli)
2. Out-of-context POS tagging
  - ▶ Annotate items in EAT and USF (stimuli and responses) with part of speech information
  - ▶ How? Most frequent POS in Web corpus ENCOW: publicly available 10-billion-word Web corpus → replicability
3. Out-of-context lemmatization
  - ▶ **morpha**, a robust morphological analyzer  
<http://users.sussex.ac.uk/~johnca/morph.html>
  - ▶ lemmatization of unknown words based on POS tag
4. Annotation with frequency information
  - ▶ frequency lists from ENCOW (lemmatised with **morpha**)

## The Free ASsociation Task (FAST) data set

### Item selection

For each stimulus in EAT (8210) and USF (5019) select a:

- ▶ **FIRST**: the most common associate response
- ▶ **HAPAX**: a response generated for the target once
  - ▶ or twice for USF (hapax responses are omitted there)
  - ▶ if several HAPAX candidates are available, pick the one whose lemma frequency matches most closely that of **FIRST**
- ▶ **RANDOM**, by randomly picking a word which was among the top 25% associates of *another stimulus* (and produced at least 5 times). If possible:
  - ▶ match lemma frequency of **RANDOM** and **FIRST**
  - ▶ try to use each **RANDOM** only once

(multiwords, numbers, closed-class words, and other words that do not occur in ENCOW were discarded)

## The FAST data set

### Final data set

- ▶ EAT subset: **3836** test items + **3774** training items
- ▶ USF subset: **2359** test items + **2360** training items
- ▶ Item = (STIMULUS, FIRST, HAPAX, RANDOM)
- ▶ Each stimulus and candidate response provided as lowercased word form and POS-disambiguated lemma
  - + ENCOW frequency information
  - + # test subjects who produced response
- ▶ Download: <https://osf.io/cd8ar/> (Evert & Lapesa 2021)
- ▶ Included as **FAST** in package **wordspaceEval**

## The FAST dataset

The new EAT task isn't perfect either ... yet

- ▶ Guessing POS from corpus doesn't always work
  - ▶ e.g. *fit*<sub>VERB</sub> → *epileptic*<sub>ADJ</sub>, *aristocracy*<sub>NOUN</sub> → *lords*<sub>NAME</sub>
  - ▶ but very few lemmatization errors (e.g. *daiquiri* → *daiquirus*)
- ▶ Colloquialisms and British slang
  - ▶ e.g. *bod*<sub>NOUN</sub> → *person*<sub>NOUN</sub> (rare in written corpus)
  - ▶ but Web corpus has Welsh *bod* 'to be' mistagged as noun
  - ▶ DSM neighbours: *yn*, *hynny*, *mewn*, *hwn*, *gyfer*, ..., 49. *bloke*, *techy*<sub>NOUN</sub>, *nus*, *hon*, ..., 60. *guy*, *mai*, *geezer*, ...
  - ▶ another example is *mellow*<sub>ADJ</sub> → *yellow*<sub>ADJ</sub>

## The FAST tasks

### Task 1: **multiple-choice**

- ▶ Given a stimulus and a  $\langle \text{FIRST}, \text{HAPAX}, \text{RANDOM} \rangle$  triple, determine which of the three candidates is FIRST.
  - ▶ Stimulus: *accept*,  $\langle \textit{receive}, \textit{love}, \textit{soul} \rangle$
- ▶ **Performance:** accuracy
- ▶ **Baseline:** 33.3%

## The FAST tasks

### Task 2: **open-vocabulary lexical access**

- ▶ Given a stimulus (e.g., *accept*), predict FIRST (*receive*) out of a candidate set (all FIRST: USF=1197, EAT=1633)
- ▶ **Performance:** two measures
  - ▶ **Soft accuracy:** average over reciprocal rank ( $1/r$ ) of the true FIRST associate, as a percentage.
    - ★ similar to accuracy of predicting first associate, but awards partial points for almost correct guesses
    - ★ always  $\geq$  top-1 accuracy
  - ▶ **Log rank:** geometric mean of  $r$  across all stimuli.
    - ★ corresponds to average over  $\log r$
    - ★ better differentiation for models that rarely get the correct answer (and hence score low on soft accuracy)
- ▶ **Baselines**
  - ▶ **Soft accuracy:** USF=0.64% and EAT=0.49%
  - ▶ **Log rank:** USF=442.0 and EAT=602.4

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FAST: Data set and tasks

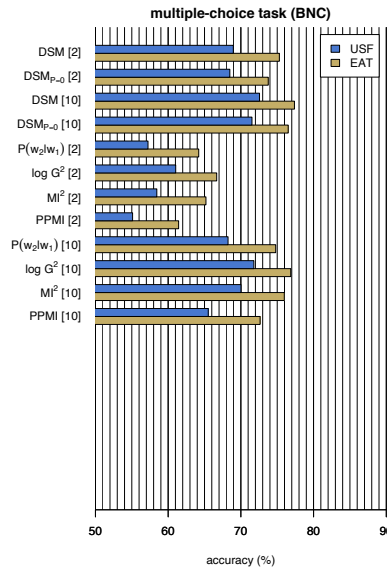
FAST: Experiments

Hands-on exercises

## Experimental setup

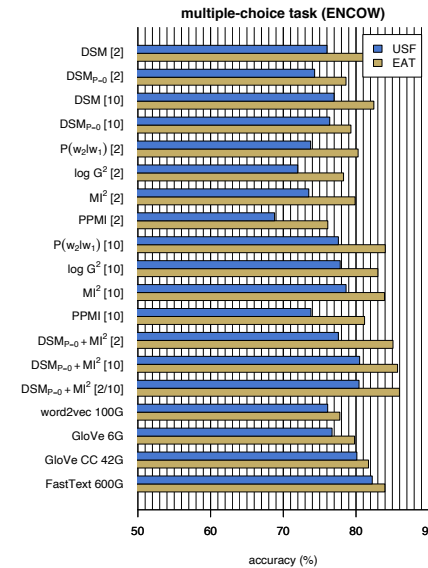
- ▶ **DSMs (second-order):** symmetric span of 2 *vs.* 10 words, other parameters set according to Lapesa & Evert (2014).
  - ▶ we experiment with Caron  $P$  (Bullinaria & Levy 2012)
  - ▶  $P = 0$  equalizes contributions of SVD dimensions
- ▶ **Collocations (first-order):** symmetric span, 2 *vs.* 10 words, with four different association measures (Evert 2008)
  - ▶ conditional probability  $P(w_2|w_1)$
  - ▶ log-likelihood  $\log G^2$  (popular for collocations)
  - ▶  $MI^2 = \log_2 \frac{O^2}{E} =$  geometric mean of  $P(w_2|w_1)$  and  $P(w_1|w_2)$
  - ▶ PPMI (popular for DSMs)
- ▶ **Corpus data:** for DSMs and collocations
  - ▶ British National Corpus: 100M words
  - ▶ ENCOW 2014 Web corpus, unique sentences: 8.5G words
- ▶ **Neural embeddings:** pre-trained models
  - ▶ word2vec (Mikolov *et al.* 2013): 100G tokens of Google News
  - ▶ GloVe (Pennington *et al.* 2014): 6G tokens Wikipedia + Gigaword
  - ▶ GloVe: 42G tokens Web data (Common Crawl)
  - ▶ FastText (Joulin *et al.* 2017): 600G tokens Common Crawl

## Results: Multiple-choice task



- ▶ British National Corpus (100M words)
- ▶ EAT vs. USF

## Results: Multiple-choice task



- ▶ ENCOW 2014 Web (8.5G words)
- ▶ EAT vs. USF
- ▶ Embeddings trained on much larger corpora
- ▶ Combined 1<sup>st</sup>-/2<sup>nd</sup>-order
  - ▶ DSM<sub>P=0</sub> + MI<sup>2</sup>
  - ▶ using neighbour rank
  - ▶ harmonic mean
  - ▶ competitive with state-of-the-art embeddings

## Results: Multiple-choice task

| model              | span | $n = 2359$ | $n = 3836$ |
|--------------------|------|------------|------------|
|                    |      | USF        | EAT        |
| DSM                | 2    | 76.01%     | 81.78%     |
| DSM <sub>P=0</sub> | 2    | 74.31%     | 78.62%     |
| DSM                | 10   | 76.98%     | 82.46%     |
| DSM <sub>P=0</sub> | 10   | 76.39%     | 79.30%     |
| $P(w_2 w_1)$       | 10   | 77.58%     | 84.02%     |
| $\log G^2$         | 10   | 77.83%     | 83.00%     |
| MI <sup>2</sup>    | 2    | 78.64%     | 83.92%     |
| PPMI               | 10   | 73.80%     | 81.18%     |
| Combined           | 2    | 77.58%     | 85.09%     |
| Combined           | 10   | 80.50%     | 85.71%     |
| Combined mix       |      | 80.41%     | 85.97%     |
| word2vec           | –    | 76.11%     | 77.78%     |
| GloVe              | –    | 76.71%     | 79.80%     |
| GloVe CC           | –    | 80.12%     | 81.72%     |
| FastText           | –    | 82.24%     | 83.97%     |

- ▶ ENCOW 2014 Web (8.5G words)
- ▶ EAT vs. USF
- ▶ Embeddings trained on much larger corpora
- ▶ Combined 1<sup>st</sup>-/2<sup>nd</sup>-order
  - ▶ DSM<sub>P=0</sub> + MI<sup>2</sup>
  - ▶ using neighbour rank
  - ▶ harmonic mean
  - ▶ competitive with state-of-the-art embeddings

## Results: Open-choice task

| model              | span | $n = 2359$ |       | $n = 3836$ |       |
|--------------------|------|------------|-------|------------|-------|
|                    |      | USF        |       | EAT        |       |
|                    |      | soft acc.  | lrank | soft acc.  | lrank |
| DSM                | 2    | 41.54%     | 6.6   | 34.53%     | 9.9   |
| DSM <sub>P=0</sub> | 2    | 42.12%     | 7.6   | 34.67%     | 12.1  |
| DSM                | 10   | 42.01%     | 6.0   | 35.93%     | 9.1   |
| DSM <sub>P=0</sub> | 10   | 42.86%     | 7.1   | 35.68%     | 11.6  |
| $P(w_2 w_1)$       | 10   | 22.34%     | 17.0  | 11.27%     | 27.1  |
| $\log G^2$         | 10   | 37.63%     | 6.6   | 34.13%     | 8.8   |
| MI <sup>2</sup>    | 10   | 39.73%     | 6.2   | 34.01%     | 8.7   |
| PPMI               | 10   | 35.34%     | 8.2   | 29.29%     | 12.2  |
| Combined           | 2    | 42.29%     | 5.5   | 37.54%     | 7.0   |
| Combined           | 10   | 44.99%     | 4.8   | 39.48%     | 6.5   |
| Combined mix       |      | 45.36%     | 4.8   | 39.48%     | 6.4   |
| word2vec           | –    | 38.98%     | 7.7   | 30.51%     | 14.8  |
| GloVe              | –    | 39.22%     | 7.6   | 30.19%     | 13.8  |
| GloVe CC           | –    | 44.01%     | 5.7   | 34.26%     | 10.5  |
| FastText           | –    | 51.00%     | 4.1   | 40.34%     | 7.2   |

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## Bonus task: Reverse free associations

The CogALex-IV shared task (Rapp & Zock 2014)

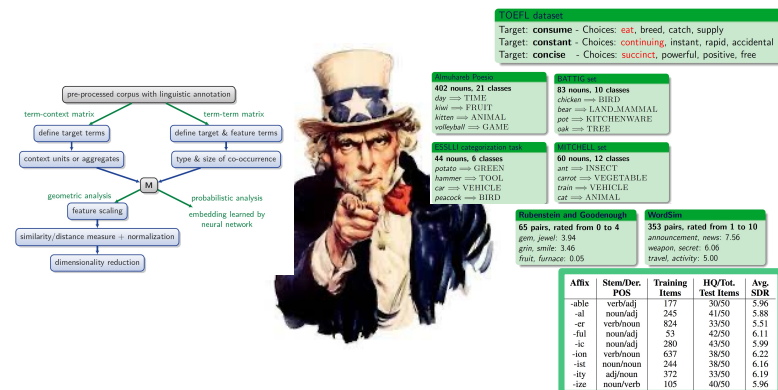
### Reverse multiword free association

- ▶ wheel, driver, bus, drive, lorry → ?
- ▶ away, minded, gone, present, ill → ?
- ▶ Data: subset of EAT (2000 stimuli each training/test)
- ▶ Very challenging (best: 35% accuracy)
  - ▶ open-ended vocabulary (including inflected surface forms!)
  - ▶ need for integrating predictions of different stimuli
- ▶ And the winner was ...
  - ▶ a system using first-order statistics to re-rank the output of a “standard” DSM (Ghosh *et al.* 2015)
  - ▶ our submission: best 1<sup>st</sup>-order: 27.7% / best 2<sup>nd</sup>-order: 14.0%
- ▶ Try it yourself: `CogALex4.rda` in `wordspaceEval`

## Hands-on exercise

- ▶ Solve the FAST multiple-choice task with a DSM
  - ▶ `eval.multiple.choice()` does most of the work for you
  - ▶ use `details=TRUE` to inspect biggest mistakes and explore performance (e.g. wrt. frequency of stimulus and response)
- ▶ Can you also make use of first-order (collocation) data?
  - ▶ hint: the DSM matrix **M** contains co-occurrence counts
- ▶ Advanced: Can you combine DSMs with first-order data?
  - ▶ hint: use average of DSM and first-order “neighbour” rank
- ▶ Advanced: Try to solve the open-choice lexical access task
  - ▶ no ready-made evaluation function in `wordspace` yet
- ▶ R code in `hands_on_part4_fast.R` will help you get started!

## It is practice session time!



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